

Exact first-order ODE

$$M(x, y) \, dx + N(x, y) \, dy = 0 \quad \text{such that} \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Exact ni?

$$4x^3 dx + 3y^2 dy = 0$$

Exact ni?

$$(2xy \, dx + dy)e^{x^2} = 0$$

Exact ni?

$$\left(\frac{3y^2 - t^2}{y^5}\right)\dot{y} + \frac{t}{2y^4} = 0$$

Ipa exact daw ni bi.

$$(6xy^3)dx + (2\alpha x^2y^2 - x \sin y)dy = 0$$

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Solution to an exact first-order ODE

$$u(x, y) = C \implies du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0$$

$$\text{By continuity: } \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

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Saon pag solve?

Ge daw bi.

$$e^{4\theta}(dr + 4r d\theta) = 0$$

Ge daw bi.

$$(2xy \, dx + dy)e^{x^2} = 0; \quad y(0) = 2$$

Some non-exact first-order ODEs can be made exact...

$$P(x, y) dx + Q(x, y) dy = 0 \quad \text{such that} \quad \frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$$

...by integrating factors

$$\mu(x, y)P(x, y) dx + \mu(x, y)Q(x, y) dy = 0 \quad \text{such that} \quad \frac{\partial(\mu P)}{\partial y} = \frac{\partial(\mu Q)}{\partial x}$$

Awa gani.

$$(x^2 + y^2) dx - 3xy dy = 0; \quad \mu = x^{-\frac{5}{3}}$$

Awa gani.

$$(3xy^3 + y^2) dx + (1 - xy) dy = 0; \quad \mu = y^{-3}$$

IF can be rare af; some heuristics:

Option 1. Assume $\mu(x, y) = x^m y^n$. Solve for m and n from exactness condition.

See <https://www.mathsisfun.com/calculus/differential-equations-exact-factors.html>.

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Option 3. Assume $\mu(x, y) = \mu(y)$. Exactness condition must be variable-separable in μ and y . Solve for $\mu(y)$.

Sulayi daw dawg.

$$(x^2 + y^2) dx = 3xy dy = 0$$

$$\mu = x^{-\frac{5}{3}}$$

$$x^{\frac{4}{3}} - 2x^{-\frac{2}{3}}y^2 = C$$

Sulayi daw dawg.

$$2x \tan y \, dx + \sec^2 y \, dy = 0$$

$$\mu = \cot y \quad x^2 + \ln|\cot y| = C$$